Step 2b - H2 - Pharma R&D faster drying

Tech baseline

Technologies used in the market

Lab convection (tray) drying oven

How it works: Forced or natural hot-air circulation around open trays; heat conducts from the surface inward.

How well it performs: Typical 2-24 h drying for wet granule/API samples at 40-80 C; units cost ~$3-15k; universal default in every lab.

Limits: Slow - heat transfer is surface-limited; thermal exposure degrades sensitive APIs; no solvent containment or recovery.

Laboratory vacuum oven

How it works: Shelf-conduction heating under reduced pressure, lowering solvent boiling point.

How well it performs: ~3-12 h cycles at 30-60 C for gram-to-kg samples; ~$5-20k; standard tool for solvent-wet pharma samples.

Limits: Conduction through a static bed keeps it at hour-scale; endpoint usually checked by opening + weighing; limited throughput for iterative R&D.

Lab fluid-bed dryer (Mini-Glatt / Caleva / Diosna class)

How it works: Heated air fluidizes a small bed of wet granules; fast convective drying.

How well it performs: 10-40 min for ~0.1-2 kg granule batches at 40-60 C; good for granule development; documented downscale behavior vs production ConsiGma-25 dryers.

Limits: Needs material that fluidizes; filter/fines issues; organic solvents require inerting which is rarely installed in labs; bigger footprint and utilities than a benchtop.

Lab freeze dryer (lyophilizer)

How it works: Frozen sample dried by sublimation under deep vacuum.

How well it performs: 24-72 h cycles; best-in-class structure/activity preservation for biologics; $20-100k+.

Limits: Very slow and energy-hungry; largely water/ice focused - irrelevant for most organic-solvent samples; the tool to beat on quality, not on speed.

Halogen/IR moisture analyzer (LOD)

How it works: Heats a small sample while weighing to determine loss-on-drying.

How well it performs: Results in ~5-15 min; $2-10k; ubiquitous for endpoint checks.

Limits: An analyzer, not a dryer - destroys the sample and cannot recover dried material for further formulation work.

Benchtop microwave-vacuum dryer (existing niche units: EnWave REV10, Kerone, Pueschner, Chinese OEMs)

How it works: Microwave field volumetrically heats solvent inside a vacuum chamber; vapour condensed separately.

How well it performs: Drying in tens of minutes at low temperature; ~9 kg water/h at 10 kW (REV10) down to ~1 kW lab units priced ~$4-6k.

Limits: Food/industrial orientation - little GxP documentation, no pharma PAT, solvent-safety and small-sample (<100 g) field-uniformity control largely unproven; fragmented low-end supply.

Why our solution can be better

For small solvent-wet samples the status quo forces hours-long vacuum-oven waits that gate formulation iteration; microwave-vacuum dries the same class of material in tens of minutes at <40-60 C, demonstrated on pharma granules in published work (~3-10x faster than convective or vacuum-only drying). Selective dielectric coupling puts energy into the solvent, not the API, so the speed gain does not trade away product quality. The advantage looks real: the gap versus existing niche MW-vacuum benchtops is a pharma-grade instrument - reproducible small-sample uniformity, solvent-safe vacuum path, GxP data, and credible residual-solvent endpoints - which is a packaging/compliance problem ProSeed can own rather than a scientific risk.

Key assumptions

• Microwave-vacuum dries 50-500 g solvent-wet granule/API samples in ~10-30 min vs 3-12 h in a vacuum oven (measured in published lab studies on pharma granules)

• Product temperature stays <40-60 C under vacuum, preserving heat-sensitive APIs (measured, MW-vacuum literature)

• Uniform fields over sub-kg samples are achievable with mode stirring/turntable; hot-spot risk manageable at small load (assumed)

• Labs will pay ~$30-80k for a GxP-capable fast drying instrument (assumed; brackets between ~$5-20k vacuum ovens and $30-100k freeze dryers)

• Ethanol/IPA/acetone are the common R&D granulation solvents, so solvent condensation and safety interlocks are mandatory, not optional (measured - standard formulation practice)

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
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Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis